

EXECUTIVE SALES DASHBOARD

Technical DAX Documentation

Microsoft Power BI Portfolio Project

Prepared by

OCHIENG' FELIX OTIENO

Bachelor of Information Technology

Data Analyst | Power BI Developer | Data Science Enthusiast

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Technical documentation of the DAX measures, business logic, and analytical calculations powering the Executive Sales Dashboard.

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1. Introduction

Introduction

Data Analysis Expressions (DAX) is the analytical engine behind the Executive Sales Dashboard. It enables the transformation of raw transactional data into meaningful business metrics through reusable calculations, time intelligence functions, and dynamic report elements.

The DAX implementation was designed with three key objectives:

- Deliver accurate business calculations.
- Improve dashboard interactivity.
- Support executive decision-making.

This document provides comprehensive technical documentation for every major measure used throughout the dashboard. Each measure is explained from both a technical and business perspective to demonstrate not only how the calculation works but also why it is valuable.

2. DAX Development Philosophy

The DAX model follows modern Business Intelligence development practices.

The guiding principles were:

- Build reusable base measures.
- Separate business logic from visual formatting.
- Use descriptive naming conventions.
- Leverage Power BI Time Intelligence functions.
- Improve readability using variables (VAR).
- Create dynamic reporting experiences.
- Optimize performance by avoiding unnecessary calculations.

These principles improve maintainability, scalability, and overall report performance.

3. Data Model Overview

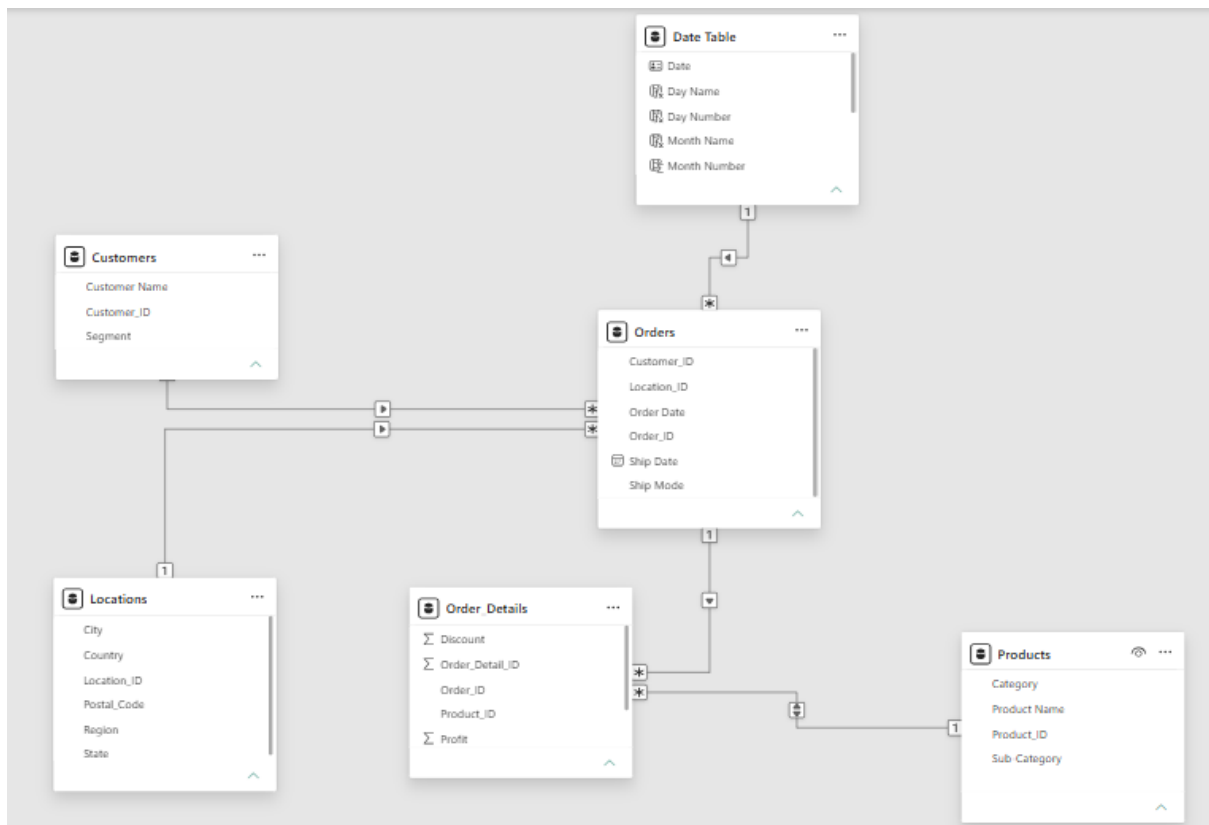


Figure 1. Star Schema Data Model

The Executive Sales Dashboard is built on a Star Schema consisting of:

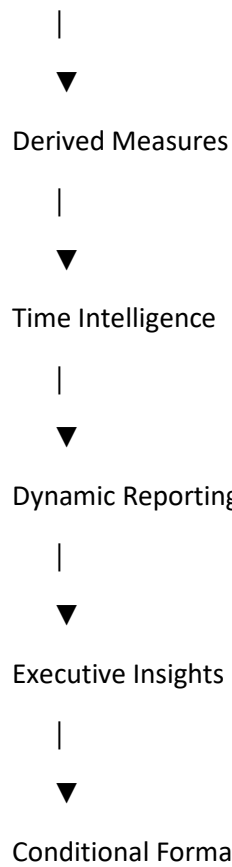
Table	Role
Orders	Fact Table
Order Details	Fact Table
Customers	Dimension
Products	Dimension
Locations	Dimension
Date Table	Calendar Dimension

The Date Table enables accurate Time Intelligence calculations, while the dimensional model supports efficient filtering and optimized query performance.

4. DAX Architecture

The measures are organized into logical layers.

Base Measures



This layered approach minimizes duplication and improves maintainability by building advanced calculations on reusable base measures.

5. Measure Categories

Category	Description
Core KPIs	Fundamental business metrics
Time Intelligence	Period-based calculations
Dynamic Reporting	Interactive titles and summaries
Executive Insights	Automated business narratives
KPI Indicators	Visual trend indicators
Conditional Formatting	Dynamic color logic

6. Measure Inventory

Measure	Category	Dashboard Usage
Total Sales	Core KPI	KPI Card
Total Profit	Core KPI	KPI Card
Total Orders	Core KPI	KPI Card
Total Quantity	Core KPI	KPI Card
Average Sales per Order	Core KPI	KPI Card
Average Profit per Order	Core KPI	KPI Card
Profit Margin	Core KPI	KPI Card
Sales MTD	Time Intelligence	KPI Card
Sales YTD	Time Intelligence	KPI Card
Previous Month Sales	Time Intelligence	Growth Analysis
Previous Year Sales	Time Intelligence	Growth Analysis
MoM Growth %	Time Intelligence	KPI Indicator
YoY Growth %	Time Intelligence	KPI Indicator
Dashboard Title	Dynamic Reporting	Dashboard Header
Mobile Dashboard Title	Dynamic Reporting	Mobile View
Analysis Title	Dynamic Reporting	Report Header
Dynamic Region	Dynamic Reporting	Dynamic Text
Dynamic Category	Dynamic Reporting	Dynamic Text
Dynamic Year	Dynamic Reporting	Dynamic Text
Selected Date Range	Dynamic Reporting	Filter Summary
Selection Summary	Dynamic Reporting	Filter Summary
Selected Region	Dynamic Reporting	Filter Summary
Executive Insight	Executive Insight	Insight Card
MoM Executive Insight	Executive Insight	Insight Card
YoY Executive Insight	Executive Insight	Insight Card

Measure	Category	Dashboard Usage
MoM Indicator	KPI Indicator	KPI Card
YoY Indicator	KPI Indicator	KPI Card
MoM Indicator Color	Conditional Formatting	Conditional Formatting
YoY Indicator Color	Conditional Formatting	Conditional Formatting

7. Detailed Measure Documentation

Total Sales

Business Purpose

Calculates the total revenue generated from all completed sales transactions.

Dashboard Usage

Executive KPI Card, trend analysis, and supporting calculations.

Business Value

Acts as the foundation for nearly all sales-related reporting and executive performance monitoring.

DAX Formula

Total Sales =

SUM(Order_Details[Sales])

Explanation

This base measure aggregates all sales values in the Order_Details fact table. It is reused throughout the model in profitability calculations, time intelligence measures, and KPI reporting, making it one of the core building blocks of the dashboard.

Total Profit

Business Purpose

Calculates the total profit generated from sales transactions.

Dashboard Usage

Executive KPI Card and profitability analysis.

DAX Formula

Total Profit =

SUM(Order_Details[Profit])

Explanation

Aggregates the profit column from the fact table and serves as the basis for profitability metrics such as Profit Margin and Average Profit per Order.

Total Orders

Business Purpose

Counts the number of unique customer orders.

Dashboard Usage

Executive KPI Card.

DAX Formula

Total Orders =

`DISTINCTCOUNT(Order_Details[Order_ID])`

Explanation

Uses DISTINCTCOUNT to ensure that each order is counted only once, even when an order contains multiple products.

Total Quantity

Business Purpose

Calculates the total quantity of items sold.

Dashboard Usage

Operational reporting and sales analysis.

DAX Formula

Total Quantity =

`SUM(Order_Details[Quantity])`

Average Sales per Order

Business Purpose

Measures the average revenue generated by each order.

Dashboard Usage

Executive performance analysis.

DAX Formula

Average Sales per Order =

DIVIDE([Total Sales], [Total Orders])

Explanation

Uses DIVIDE() to safely calculate average order value while avoiding divide-by-zero errors.

Average Profit per Order

Business Purpose

Calculates the average profit generated by each order.

DAX Formula

Average Profit per Order =

DIVIDE([Total Profit], [Total Orders])

Profit Margin

Business Purpose

Measures overall profitability as a percentage of sales.

DAX Formula

Profit Margin =

DIVIDE([Total Profit], [Total Sales])

Business Value

Provides executives with an efficiency metric that complements revenue reporting.

Time Intelligence Measures

The dashboard includes several advanced time-based measures:

- Sales MTD
- Sales YTD
- Previous Month Sales
- Previous Year Sales
- MoM Growth %
- YoY Growth %

These measures enable comparisons across reporting periods and support trend analysis.

Dynamic Reporting Measures

The dashboard uses DAX to create a more interactive reporting experience.

- Dashboard Title
- Mobile Dashboard Title
- Analysis Title
- Dynamic Region
- Dynamic Category
- Dynamic Year
- Selected Date Range
- Selected Region
- Selection Summary

Executive Insight Measures

The following measures translate numerical performance into business-friendly narratives:

- Executive Insight
- MoM Executive Insight
- YoY Executive Insight

These measures use SWITCH() and conditional logic to generate contextual messages based on business performance.

KPI Indicators and Conditional Formatting

Document:

- MoM Indicator
- YoY Indicator
- MoM Indicator Color
- YoY Indicator Color

8. DAX Best Practices

The following practices were applied throughout the project:

Reusable Measures

Base measures such as **Total Sales** and **Total Profit** are referenced by other calculations, reducing duplication.

Safe Division

DIVIDE() is used instead of the / operator to gracefully handle division-by-zero scenarios.

Time Intelligence

A dedicated Date Table supports accurate MTD, YTD, MoM, and YoY calculations.

Dynamic Reporting

Measures drive titles, filter summaries, and executive insights, creating a responsive reporting experience.

Naming Conventions

Measures use descriptive names that improve readability and simplify maintenance.

Conditional Formatting

Presentation logic is separated into dedicated color measures, improving maintainability and consistency.

9. Performance Optimization

Several design decisions were made to improve report performance:

- Reuse of base measures.
- Star Schema data model.
- Dedicated Date Table.
- Optimized relationships.
- Efficient filter propagation.
- Limited use of complex iterator functions.
- Clear separation of calculation and formatting logic.

These practices contribute to a responsive and scalable dashboard.

10. Business Impact

The DAX implementation enables the dashboard to:

- Calculate executive KPIs accurately.
- Support advanced time-based analysis.
- Provide dynamic report elements.
- Generate automated business narratives.

- Enhance user interaction.
- Improve executive decision-making.

The measures transform raw transactional data into actionable insights while maintaining performance and maintainability.

11. Conclusion

The DAX measures developed for the Executive Sales Dashboard form the analytical foundation of the solution. By combining reusable calculations, advanced time intelligence, dynamic reporting, and business-focused insights, the model supports accurate reporting and informed decision-making.

The implementation reflects industry best practices in Power BI development and demonstrates the practical application of DAX in solving real-world business reporting challenges. Together with the Star Schema data model and interactive dashboard design, these measures create a scalable, maintainable, and executive-ready Business Intelligence solution.